

Budget — AKG-E Full Proposal (DRAFT)

(DRAFT — for PI / lab-admin review; dollar amounts are placeholders)

Project: Agentic Knowledge Graphs for AI-Accelerated Engineering with Simulation-Grounded Validation

PI: Kereshmeh Afsari **co-PI:** Jason Cusati

Sponsor: Amazon–Virginia Tech Initiative, CFP 2026–2027

Period: 12 months (3-month aggressive build + 9-month dissemination / scaling)

Category	Amount
Personnel	
GRA #1 — Cusati (12 mo; full-time during 3-mo build)	\$XX,XXX
Stipend	\$XX,XXX
Tuition	\$XX,XXX
Benefits / fringe	\$X,XXX
GRA #2 — TBD (12 mo; full-time during 3-mo build)	\$XX,XXX
Stipend	\$XX,XXX
Tuition	\$XX,XXX
Benefits / fringe	\$X,XXX
Faculty summer salary — Afsari, 1 mo (optional)	\$XX,XXX
Materials and Computing	
LLM API tokens (development + benchmark runs)	\$XX,XXX
AWS Bedrock + cloud compute (training / serving / KG)	\$XX,XXX
Simulation software / FEA licenses	\$X,XXX
Travel	
Conference venue #1 (AI for engineering / agentic AI)	\$X,XXX
Conference venue #2 (construction / civil engineering)	\$X,XXX
Amazon collaboration travel (Seattle / Arlington)	\$X,XXX
Total Direct Costs	\$XXX,XXX
Indirect Costs (VT F&A rate, to be confirmed by lab admin)	\$XX,XXX
Total Project Cost	\$XXX,XXX

Justification — LLM API Tokens. AKG-E is, by construction, a high-volume LLM workload: each iteration of the propose → VVUQ → feedback loop spends tokens on retrieval-augmented generation over the curated knowledge graph, on structured graph queries that surface code provisions and physical-law constraints, on simulation feedback ingestion (FEA / digital-twin / PDE solver outputs returned to the agent), and on the refinement step that consumes gate-emitted violations to produce the next candidate artifact. The benchmark evaluation amplifies this footprint substantially: three baselines (B0 Claude alone; B1 Claude + graph-RAG; B2 full AKG-E) are run across three engineering domains (construction, transportation, energy), with $n \geq 5$ random seeds per cell and bootstrap resampling for 95% confidence intervals on the headline correctness, calibration, and efficiency metrics. The full benchmark therefore drives thousands of multi-turn agent runs at production-quality model tokens, and the LLM API line is sized to support both day-to-day development and full benchmark execution without throttling. Separating this line from cloud compute prevents the token spend from being absorbed into a generic AWS line item where Amazon reviewers and VT lab admin cannot audit it independently.

Notes for reviewers: All dollar figures above are placeholder “XX,XXX” tokens pending PI sign-off and VT lab-admin confirmation of (a) GRA stipend and tuition rates for the 2026–2027 academic year, (b) the VT F&A indirect-cost rate applicable to this sponsor, and (c) the faculty-summer-salary decision (included as an optional line; the PI may zero it). The total ask will be finalized within the Amazon-VT CFP Phase 1 envelope.